
Project Based Learning

Project Title - A Web Platform for Student to Student Trade

**Project Domain : Web Application
Development**

Group Id : E9-03

Team Members :

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Problem Statement -

- College students buy academic materials for short-term use, leading to unused resources after each semester
- Other students must repurchase the same materials at high cost
- Informal resale methods are unorganized and unreliable
- Existing platforms are not student-verified or campus-specific
- There is no structured system focused on academic resource reuse.

Some of the most common issues include

In colleges, academic materials such as textbooks, lab manuals, reference books, calculators, and notes are often required only for a single semester. After use, these resources remain unused, while incoming students are forced to buy the same materials again at high prices. The absence of a structured, campus-specific platform results in wastage, financial burden, and inefficient resource circulation.

**Short-Term
Usage, Long-
Term Waste**

**High Cost for
Students**

**Unorganized
Resale
Methods**

**No Campus-
Focused
Solution**

Motivation & Need of the Project

Why We Chose This Problem

This problem is directly experienced by college students every semester. Being students ourselves, we observed repeated purchases, wasted materials, and lack of an efficient reuse system within campuses.

Relevance to Campus / Local Area

Addresses a common campus issue by creating a trusted, student-verified system for sharing and reusing academic resources locally

Relevance to Society & Industry

Promotes sustainable consumption by reducing academic waste. Encourages circular economy practices and cost-effective resource utilization in the education sector.

Role of Emerging Technology & Impact

Uses digital platforms for verification, listings, and secure exchanges. Helps students save money, reduces waste, and creates an organized academic reuse ecosystem.

Scope of the Mini Project

In-Scope Features

- ✔ Student-verified platform for a single campus
- ✔ Listing of academic materials (books, notes, lab manuals)
- ✔ Semester-wise and category-based browsing
- ✔ Basic search functionality for resources
- ✔ Direct student-to-student interaction
- ✔ Real-time status updates for availability

Out-of-Scope Features

- ✘ Online payment or transaction processing
- ✘ Delivery or logistics management
- ✘ Cross-college or public marketplace access
- ✘ AI-based recommendations or analytics
- ✘ Native mobile app development (iOS/Android)
- ✘ Legal dispute handling or refund mechanisms

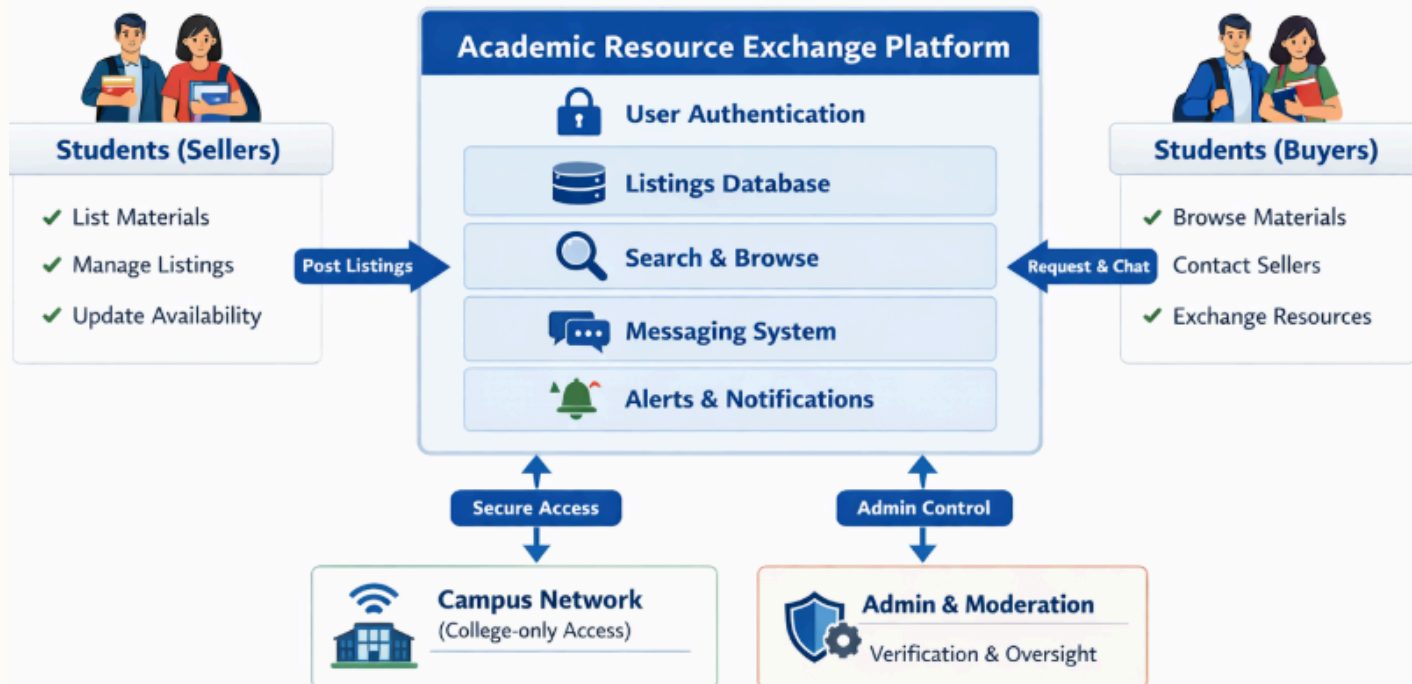
Proposed Solution / Approach

High-Level Idea

A centralized, campus-specific platform enabling the seamless reuse of academic resources. By connecting students, we promote cost-effective and sustainable sharing.

System Functionality

Registration via verified credentials allows users to list materials post-semester. Interested students can browse and initiate contact within the community.



Technology Stack

Backend / Application Logic (Node & Express – MERN)

- Node.js
- Express.js
- RESTful Architecture

Frontend Technologies (React – MERN)

- React.js
- HTML5
- CSS3
- JavaScript (ES6+)

Database (MongoDB – MERN)

- MongoDB

APIs / Frameworks

- REST APIs
- JWT (JSON Web Tokens)
- Axios / Fetch API

Tools & Platforms

- Visual Studio Code
- Git
- GitHub
- Postman
- AWS / Render / Firebase

Literature Survey / Existing System

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1. OLX

- Online platforms for buying and selling second-hand items
- Mainly focused on general public and commercial transactions

Limitations:

- **No college-specific filtering**
- **Trust and authenticity issues**
- **Not suitable for academic materials like notes or lab equipment**



2. Facebook Marketplace / WhatsApp Groups

- Informal peer-to-peer exchange through social media

Limitations:

- **Unorganized listings**
- **No proper search or categorization**
- **No verification of users**
- **Messages and data get lost easily**

The End

Thank you for listening